



The
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Advanced Encryption Standard

- The more popular and widely adopted symmetric encryption algorithm likely to be encountered nowadays is the *Advanced Encryption Standard (AES)*.
- It is found at least six times faster than triple DES.
- A replacement for DES was needed as its key size was too small.
- With increasing computing power, it was considered vulnerable against exhaustive key search attack.
- Triple DES was designed to overcome this drawback but it was found slow.

The features of AES

- Symmetric key symmetric block cipher
- Block size = 128 bits
- 128-bit data, 128/192/256-bit keys
- Stronger and faster than Triple-DES
- Provide full specification and design details
- Software implementable in C and Java

Operation of AES

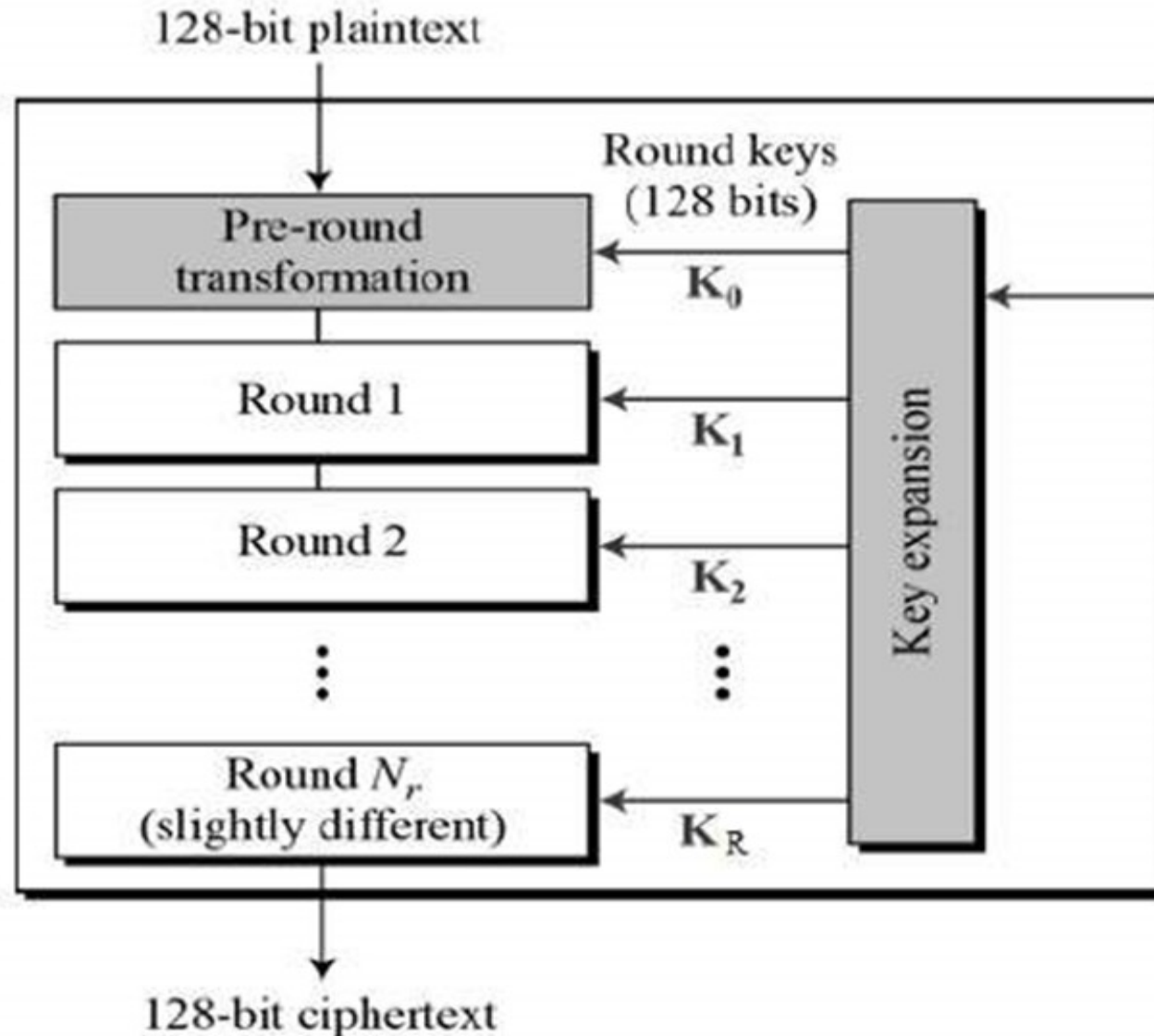
- AES is an iterative rather than Feistel cipher.
- It is based on 'substitution–permutation network.
- It comprises of a series of linked operations, some of which involve replacing inputs by specific outputs (*substitutions*) and others involve shuffling bits around (*permutations*).

Cont.

- Interestingly, AES performs all its computations on bytes rather than bits.
- Hence, AES treats the 128 bits of a plaintext block as 16 bytes.
- These 16 bytes are arranged in four columns and four rows for processing as a matrix .
- Unlike DES, the number of rounds in AES is variable and depends on the length of the key.
- ***AES uses 10 rounds for 128-bit keys, 12 rounds for 192-bit keys and 14 rounds for 256-bit keys.***
- Each of these rounds uses a different 128-bit round key, which is calculated from the original AES key.

State Matrix

Each block represents
1-byte



R	Key size
10	128
12	192
14	256

Relationship between
number of rounds(R)
and cipher key size

State Matrix

Key Size= 128-Bits or 4 Words

K0	K4	K8	K12
K1	K5	K9	K13
K2	K6	K10	K14
K3	K7	K11	K15

Key Expansion
Algorithm

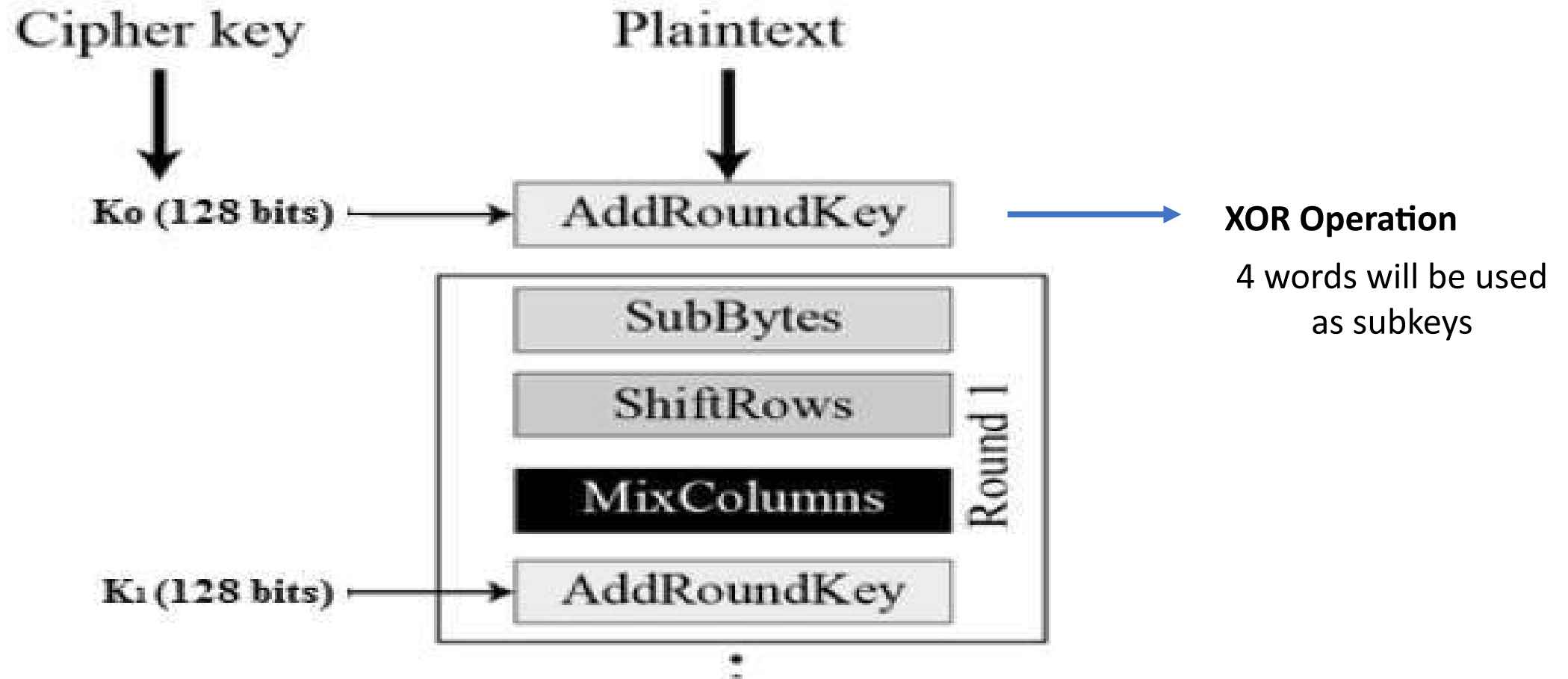
K0	K1	K2	...	K41	K42	K43
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44 Words

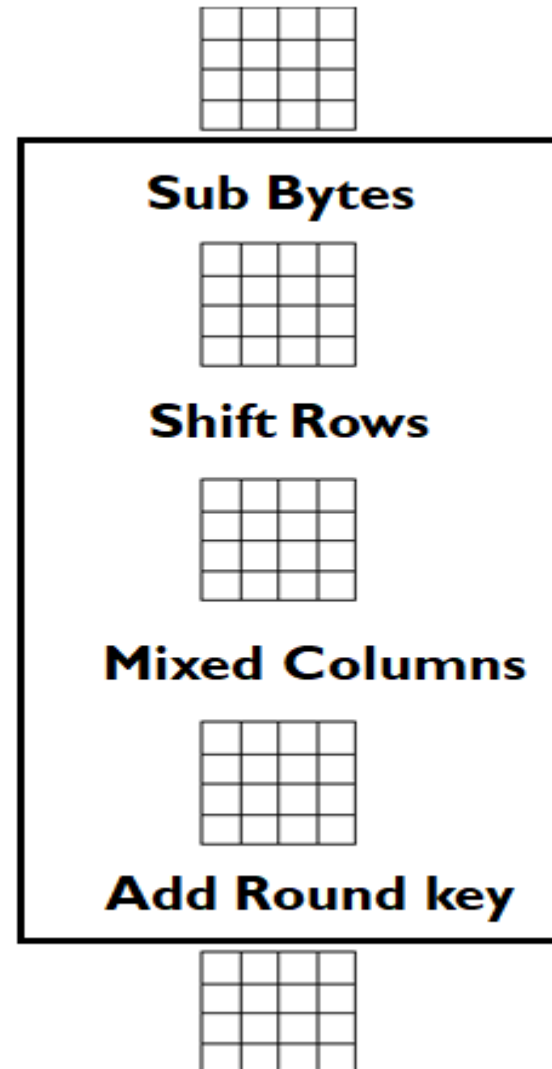
S0,0	S0,1	S0,2	S0,3
S1,0	S1,1	S1,2	S1,3
S2,0	S2,1	S2,2	S2,3
S3,0	S3,1	S3,2	S3,3

[W0, W1, W2, W3]

Encryption Process



Structure of each round

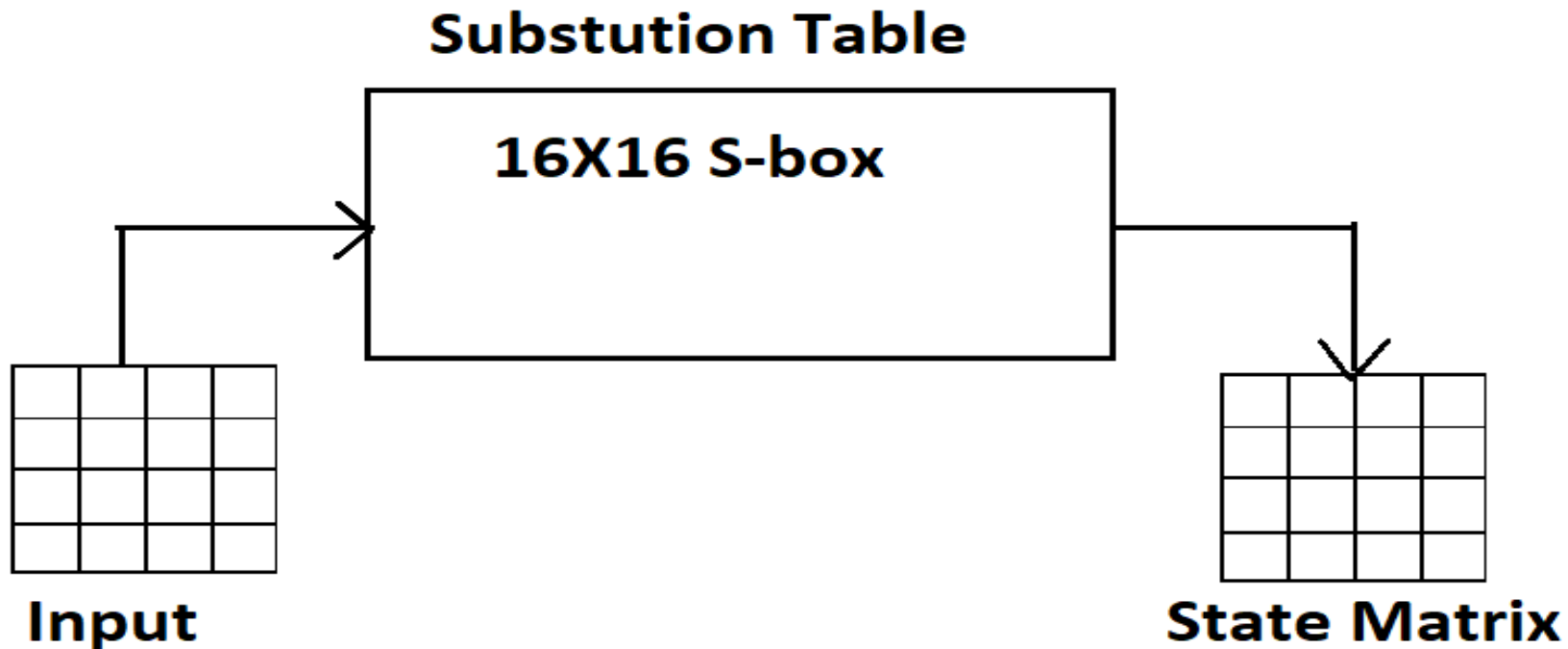


- There are four types of transformation
 - Substitution
 - Permutation
 - Mixing
 - Key-adding

Cont.

Byte Substitution (SubBytes)

- The 16 input bytes are substituted by looking up a fixed table (S-box) given in design.
- The result is in a matrix of four rows and four columns.

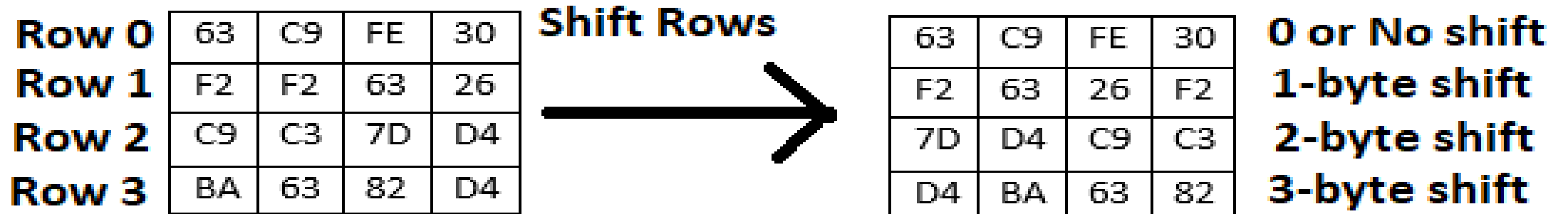


	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
a	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
b	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
c	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
d	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
e	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
f	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

Only one table is used for the transformation of the bytes, which means if two bytes are same then the transformation will also be the same.

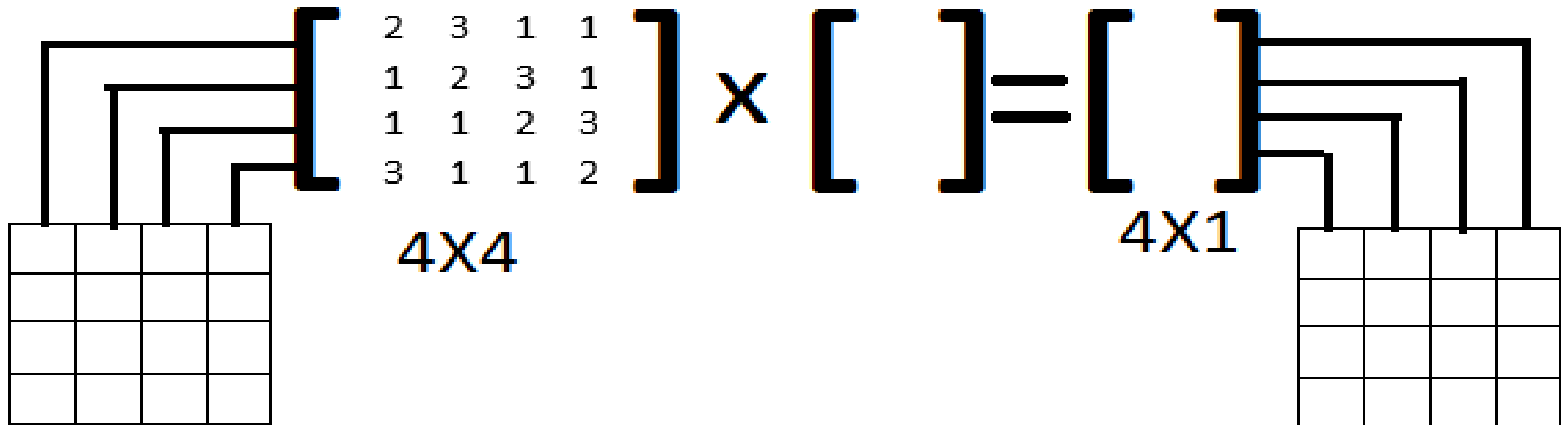
Shift-Rows/Permutation

- Each of the four rows of the matrix is shifted to the left.
- Any entries that 'fall off' are re-inserted on the right side of row.



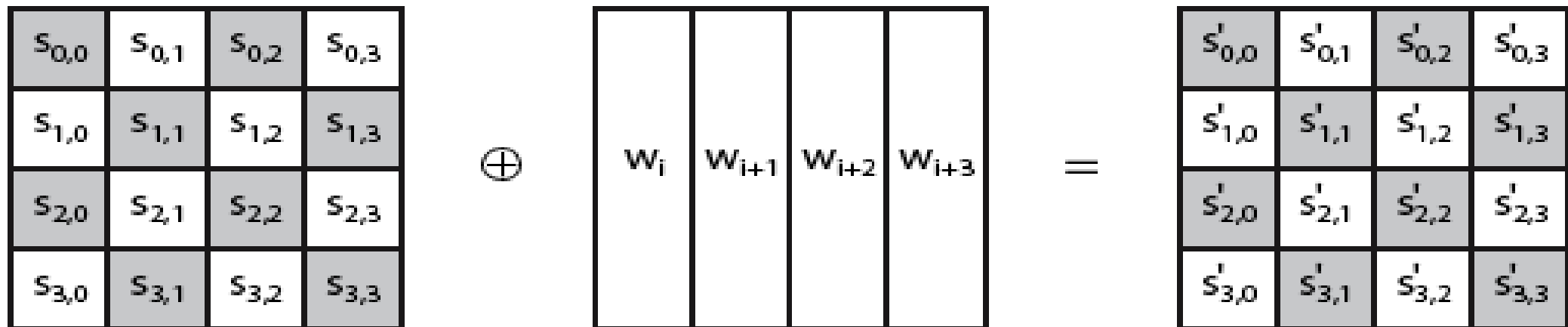
Mix Columns

- Each column of four bytes is now transformed using a special mathematical function.
- This function takes as input the four bytes of one column and outputs four completely new bytes, which replace the original column.
- The result is another new matrix consisting of 16 new bytes.
- It should be noted that this step is not performed in the last round.



Add round key

- The 16 bytes of the matrix are now considered as 128 bits and are XORed to the 128 bits of the round key.
- If this is the last round then the output is the cipher-text.
- Otherwise, the resulting 128 bits are interpreted as 16 bytes and we begin another similar round.



Question?